Homework 10 (MIP and Duality)

April 19, 2026

This is the last homework assignment. Problems 1-3 are each worth 10 points and problem 4 is worth 40 points. **Please submit your code outputs to Crowdmark. In addition, please submit your code as a .ipynb file or screenshots of all code used to solve the problems.**

1)

Consider the following Integer Program (IP):

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} \leq \vec{b}$$

$$0 \leq \vec{x} \leq \vec{d}$$

and

$$\vec{x} \in \mathbb{Z}^n$$

where $\vec{c}, \vec{d} \in \mathbb{R}^n$, $\vec{b} \in \mathbb{R}^m$ such that $\vec{d} \geq 0$.

a)

Show that this IP can be reformulated as a Binary Integer Program (BIP)

b)

Can every IP be reformulated as a BIP? Provide a proof/disproof for your answer.

2)

Convert the following LPs into their respective dual LPs.

a)

$$\text{minimize } 3x_1 + 2x_2 - 3x_3$$

s.t.

$$x_1 - x_2 + x_3 \leq 4$$

$$x_1 + x_2 - x_3 \leq 6$$

$$x_1 + 2x_2 - 3x_3 = 1$$

$$x_1, x_2 \geq 0,\ x_3 \in \mathbb{R}$$

b)

$$\text{minimize } x_1 - 2x_2 + x_3$$

s.t.

$$x_1 + x_2 - 2x_3 \geq 4$$

$$x_1 - x_2 - x_3 \leq 2$$

$$x_1 + 2x_2 - 3x_3 \geq 1$$

$$x_1 \leq 0 \; x_2, x_3 \in \mathbb{R}$$

3)

Let A be a symmetric square matrix. Consider the LP below:

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} \geq \vec{c}$$

$$\vec{x} \geq 0$$

Prove that if $\vec{x}^*$ satisfies $A\vec{x}^* = \vec{c}$ and $\vec{x}^* \geq 0$, then $\vec{x}^*$ is an optimal solution.

4)

In the video game, Pokemon Diamond, you are a Pokemon trainer who has the goal of creating the best Pokemon team. At the end of the game, you must travel through a difficult path called, "Victory Road". Using your team of 6 Pokemon, you need to battle difficult trainers and must solve puzzles using special moves called field moves. Each Pokemon has one or two types. The primary and secondary type are treated the same mechanically in the game. Pokemon are also able to learn various moves, including field moves. Each Pokemon can only learn certain moves. Some Pokemon are incapable of learning certain moves. In addition, each Pokemon has a base stat total. This number corresponds to the effectiveness of this Pokemon in a battle. A higher base stat total tends to suggest a stronger Pokemon. Suppose you want to create a team from a collection of 30 Pokemon with the goal of getting through victory road. This team should have the maximum possible base stat total subject to the following rules:

1. The team cannot consist of more than 6 Pokemon.
2. There should be at least one Pokemon on the team who knows each of the required field moves. These required moves are "Rock Smash", "Surf", "Strength", "Rock Climb", and "Waterfall".
3. No Pokemon on the team can learn more than 2 field moves.
4. There cannot be more than 1 Pokemon on the team with a given type (this type can be primary or secondary. For example a Fire/Ghost and a Ghost/Psychic Pokemon will each share the Ghost type, so we cannot include both on the team).

The specific allowed Pokemon as well as their types, base stats, and ability to learn each field move are contained in the "Pokemon_Problem.csv" file. The top row contains a header row, and the rows below contain the Pokemon names, types, ability to learn a given field move (1 if the Pokemon can learn it, 0 otherwise), and base stat total. If a pokemon has only one type, the secondary type column will contain a "None" entry. All relevant information comes from https://www.smogon.com/dex/dp/Pokemon/. All Pokemon copyrights are held by Nintendo (please don't sue us Nintendo).

a)

Formulate the above problem as an MIP.

b)

Plug this into a Python solver and determine an optimal team. Be sure the list the Pokemon on the team in your final answer (You might find it helpful to use Pandas or a related library in Python to read in the relevant .csv data). Python may convert the "None" entries to "NaN" when reading in the .csv file. Be sure to clean up the .csv as needed.